

Trust in Autonomous Human-Robot Collaboration: Effects of Responsive Interaction Policies

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I. INTRODUCTION

Trust is essential for effective human-robot collaboration (HRC). Most prior HRI research uses Wizard-of-Oz control or scripted interactions — shielding participants from the errors and breakdowns that define real autonomous systems. Speech recognition failures, delayed responses, and misinterpretations are not peripheral issues: they are defining features of deployed robots, and likely shape trust in ways that scripted studies cannot reveal.



Figure 1: Experimental setup: Misty II social robot engaged in autonomous dialogue with a participant.

This pilot study addresses that gap by examining how interaction policy shapes trust under fully autonomous, affect-adaptive spoken dialogue using the Misty II social robot. We compared a proactive, affect-responsive policy — one that adapted its assistance and dialogue based on inferred user state — against a reactive, task-focused policy that responded only when prompted. Because all dialogue management operated without human intervention, participants were exposed to the real latency, recognition errors, and breakdowns that scripted studies conceal.

II. METHOD

Participants: N = 29 recruited; 24 eligible after exclusions for sustained communication failure

Robot Platform: Misty II social robot

- Real-time pipeline: speech recognition → emotion inference (DistilRoBERTa) → affect-adaptive dialogue generation (Gemini) → embodied robot response
- No human intervention during sessions

Design: Between-subjects

- Responsive Policy:** Proactive, affect-adaptive assistance — empathy, encouragement, and collaborative language based on inferred interaction state
- Control Policy:** Reactive, task-focused responses only; assistance provided only when explicitly requested

Collaborative Tasks:

- Robot-dependent suspect identification (5 min, enforced collaboration)
- Optional-collaboration location puzzle (10 min, voluntary collaboration)

Measures:

- Trust in Industrial HRC (TI-HRC) — experienced/affective trust formed during interaction

- Trust Perception Scale-HRI (TPS-HRI) — retrospective perceived reliability
- Task accuracy
- Dialogue coding (interaction quality, communication breakdown rate)

III. PRIMARY ANALYSIS

i. Responsive Policy Increased Trust (eligible sample, n = 24)

Responsive vs. Control (eligible sample, n = 24):

- +26 points on Experienced Trust** (TI-HRC), $p = .004$ (affective, embodied trust formed during interaction)
- +15 points on Perceived Trust** (TPS-HRI), $p = .022$ (retrospective reliability judgements)
- Both effects held after controlling for baseline robot attitudes (NARS)
- Effects were **independent of task accuracy** ($p = .47$)

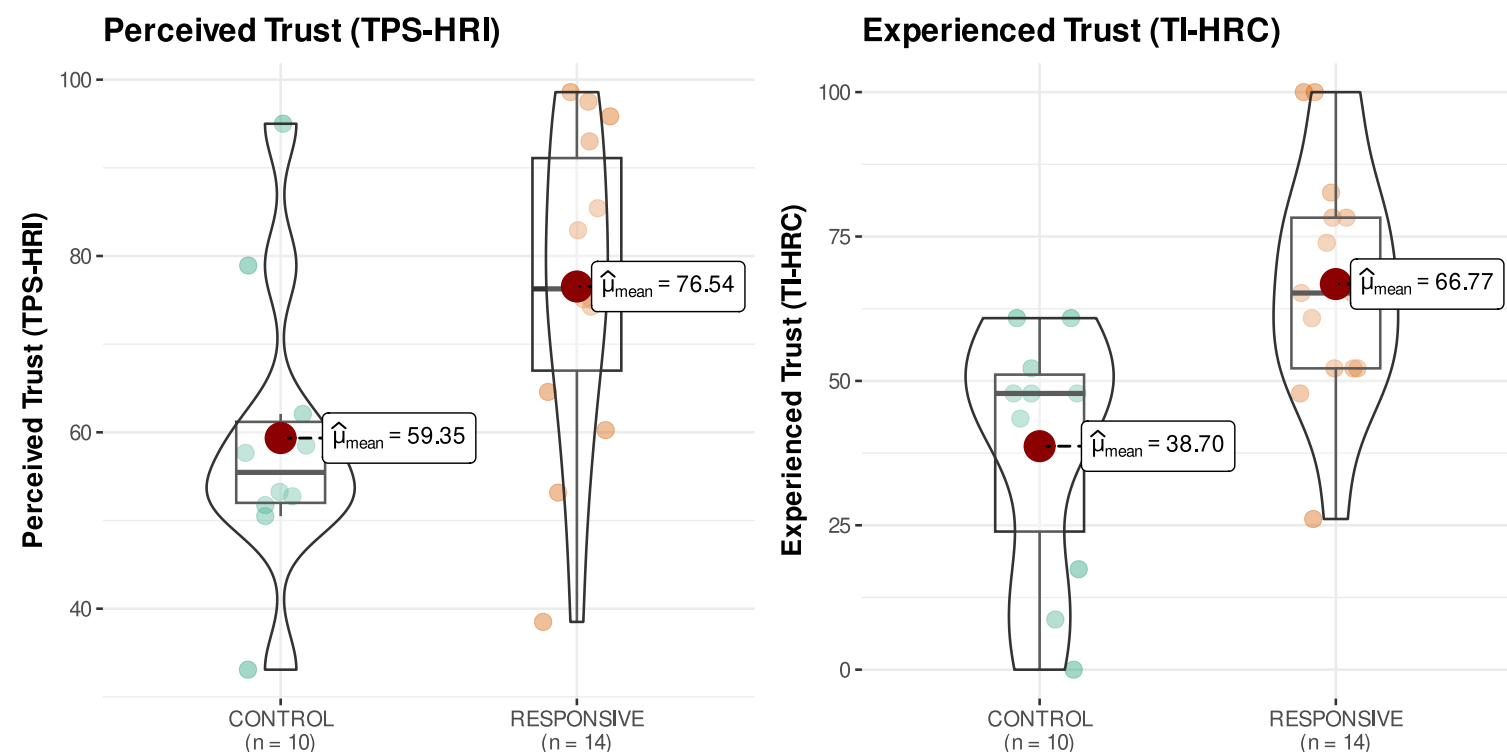


Figure 2: Distribution of Trust Perception by interaction policy. Points represent individual observations; violins depict score distributions. Red points indicate group means with 95% confidence intervals. Statistical comparisons are reported in the Results section.

ii. Task Performance: No Significant Difference

Overall task accuracy did not differ between conditions (60% vs. 66%, $p = .47$), suggesting trust was shaped by interaction quality, not task outcomes. While the robot-dependent task showed a descriptive advantage for the Responsive condition (64% vs. 30%) it did not reach significance at this sample size.

Characteristic	CONTROL (N = 10)	RESPONSIVE (N = 14)	p-value
Experienced Trust (TI-HRC)	39 (22)	67 (21)	0.004
Perceived Trust (TPS-HRI)	59 (17)	77 (18)	0.022
Overall Task Accuracy	0.60 (0.21)	0.66 (0.23)	0.47

IV. EXPLORATORY MECHANISM ANALYSIS

i. Communication Viability as a Boundary Condition (full sample, n = 29)

Despite specifying English fluency as an eligibility criterion, substantial variability in spoken-language proficiency emerged during in-person sessions—a constraint that a Wizard-of-Oz design would have silently compensated for. Five sessions were classified as non-viable based on dialogue coding (>60% of turns affected by ASR failure). This unplanned finding motivated exploratory sensitivity analyses examining how communication breakdown moderated policy effects.

Breakdown affected the two trust dimensions differently:

- Experienced trust (TI-HRC):** responsiveness advantage attenuated progressively as breakdown accumulated — affective trust depends on sustained interaction quality
- Perceived reliability (TPS-HRI):** more robust to breakdown — evaluative judgements appeared to anchor on discrete successful exchanges

When communication collapsed entirely, trust ratings became difficult to interpret as calibrated evaluations of collaborative competence.

Trust as a Function of Communication Breakdown

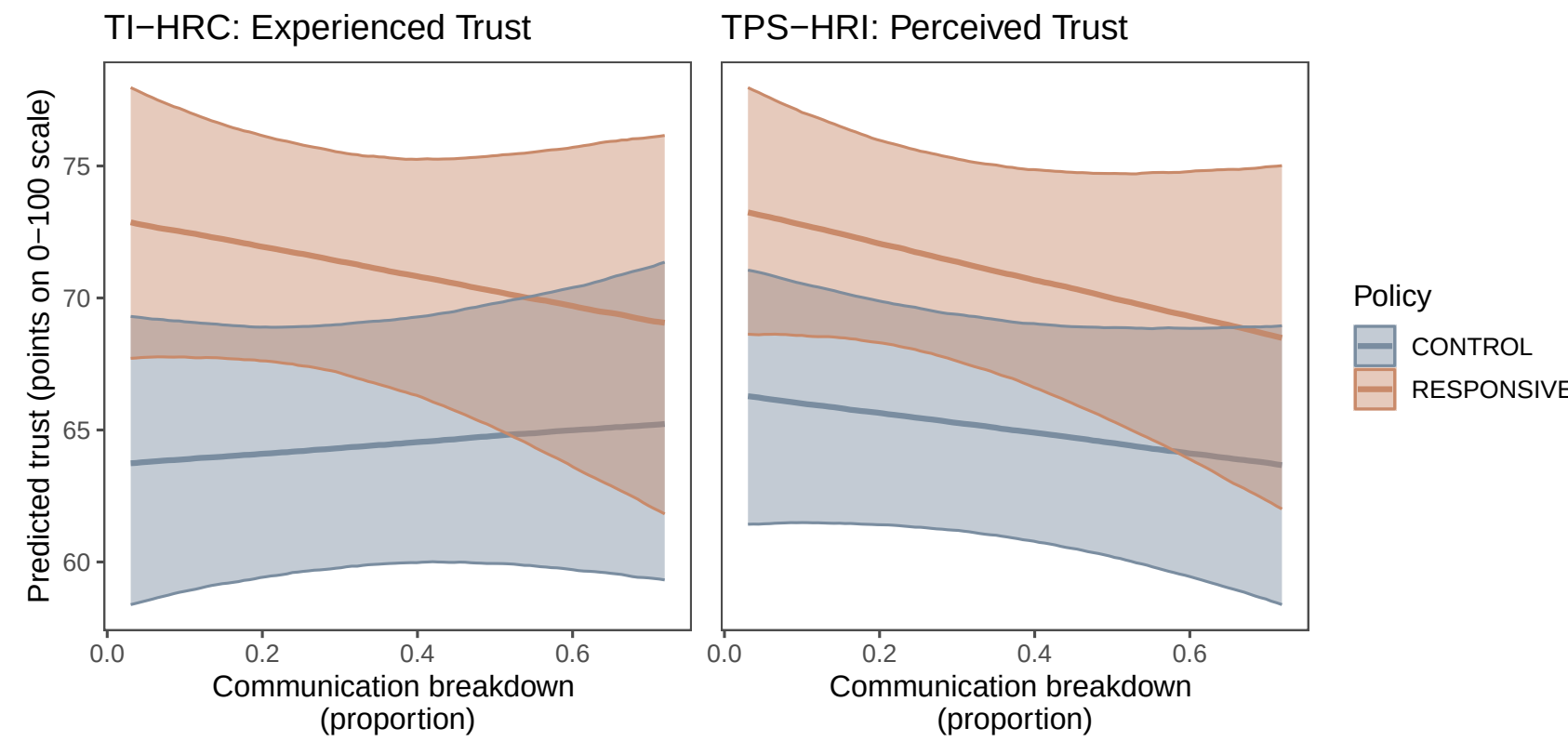


Figure 3: Experienced trust (TI-HRC) showed a clearer responsiveness advantage under low breakdown that attenuated as communication failed; perceived reliability (TPS-HRI) was less sensitive to breakdown across both conditions.

The apparent uptick in control condition experienced trust at higher breakdown levels may reflect noise given sparse data at that range, reduced cognitive load from minimal interaction, or comprehension limitations affecting post-test responses — these interpretations remain open.

V. TAKEAWAYS

- Robots that proactively adapt to users — offering unprompted help and empathy — are rated more trustworthy than robots that only react, independent of task accuracy
- Affective trust broke down faster than perceived reliability as communication failed, suggesting these dimensions reflect distinct aspects of collaborative experience
- Trust advantages of responsive robots disappear when communication breaks down, making viability a prerequisite not just a moderator
- Studying trust under real autonomy exposes constraints that Wizard-of-Oz designs conceal and that matter for deployment

VI. LIMITATIONS

This pilot used a modest eligible sample (n = 24) with semi-random condition assignment, limiting statistical power for higher-order effects. In addition, the responsive policy represents a bundled interaction profile; future work should disentangle affect-adaptation from increased dialogue exposure and session duration. Spoken-language proficiency variability among participants produced non-viable sessions that required exclusion, which may have introduced selection bias. This highlights the need for adaptive policies that detect emerging breakdown rather than assuming linguistic competence. These findings are exploratory and intended to inform a larger confirmatory study.

Full analytic details and sensitivity analyses available at arXiv link below.